Supplemental Material: Annu. Rev. Econ. 2014 . 6:445–74
doi: 10.1146/annurev-economics-080213-040918
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Appendix to: Optimal Design of Funded Pension Schemes

Lans Bovenberg* Roel Mehlkopf†

November 10, 2013

Abstract: This Appendix derives the analytical expressions in the above mentioned paper.

*Tilburg University and Netspar, P.O. Box 90153, 5000 LE Tilburg, The Netherlands. E-mail: A.L.Bovenberg@tilburguniversity.edu

†Tilburg University and Netspar, P.O. Box 90153, 5000 LE Tilburg, The Netherlands. E-mail: R.J.Mehlkopf@tilburguniversity.edu

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Appendix

Technical derivation of expressions in chapter 2

Optimization of individual decision making

In order to come to the optimal investment and consumption strategies in the life-cycle model of chapter 2, we proceed backwards. The Hamilton-Jacobi-Bellman equation for the problem is given by, with $J(W,t)$ denoting the value function:

$$\rho J = \max_{x,C} \left(\frac{1}{1-\theta} C^{1-\theta} + J_F(F(f^*\sigma\lambda + r) + 1 - C) + \frac{1}{2} J_{FF}(F f^* \sigma)^2 + J_t \right). \tag{1}$$

where the subscripts of the value function J denote first- and second order derivatives with respect to financial wealth F_t. We know that the value function is of the form: (see e.g. Bodie Z, Merton RC, Samuelson WF.1992. Labor Supply Flexibility and Portfolio Choice in a Life-Cycle Model. *Journal of Economic Dynamics and Control*. 16: 427-449.)

$$J(F,t) = \frac{(F_t + H_t)^{1-\theta}}{1-\theta} g(D-t)^{\theta}, \tag{2}$$

where $g(\cdot)$ is a funtion of time and where H_t denotes the human capital of the investor at time t. We have for $t \in [0,T]$:

$$H_t = \frac{1}{r}\left(1 - e^{-r(T-t)}\right) \tag{3}$$

and $H_t = 0$ for $t \in (T,D]$. Substituting of the value function and solving for the resulting first order conditions in (1), we find for the optimal consumption and portfolio policy:

$$C_t = (F_t + H_t) g (D-t)^{-1}, \tag{4}$$

$$f_t^* = \frac{1}{\theta}\frac{\lambda}{\sigma}\left(1 + \frac{H_t}{F_t}\right). \tag{5}$$

Let $W_t \equiv F_t + H_t$ denote the total wealth of the investor while f_t denotes the fraction of total wealth invested in risk bearing assets. From this it follows that $f_t^* = f_t(1 + \frac{H_t^i}{F_t^i})$. The results in equations (6) and (7) now directly follow

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from equation (5). Substitution of the optimal policies (4) and (5) into (1) provides a linear ordinary differential equation for the function $g(D-t)$ which is solved for:

$$g(D-t) = \frac{1}{A}(e^{A(D-t)} - 1), \tag{6}$$

where

$$A = \frac{(1-\theta)r - \rho}{\theta} + \frac{1}{2}\frac{1-\theta}{\theta^2}\lambda^2. \tag{7}$$

Derivation of consumption dynamics

Applying Ito's lemma to (4), we find that the consumption dynamics are given by:

$$dC_t = g(D-t)^{-1}dW_t - \frac{\dot{g}(D-t)(W_t)}{g(D-t)^2}dt. \tag{8}$$

where $\dot{g}(\cdot)$ denotes the first-order derivative of function $g(\cdot)$. Employing the optimal portfolio rule given in (5), we find for the wealth dynamics:

$$\frac{dW_t}{W_t} = \left(\frac{\lambda^2}{\theta} + r - g(D-t)^{-1}\right)dt + \frac{\lambda}{\theta}dZ_t. \tag{9}$$

Using the expression for $g(D-t)$ in (7), we have:

$$-g(D-t)^{-1} - \frac{\dot{g}(D-t)}{g(D-t)} = A. \tag{10}$$

Substituting this result into of (10) and (9) into (8), we establish:

$$\frac{dC_t}{C_t} = \left(\frac{\lambda^2}{\theta} + r + A\right)dt + \frac{\lambda}{\theta}dZ_t. \tag{11}$$

Substitution of (7) and taking expectations on both sides yields

$$E_t\frac{dC_t}{C_t} = \frac{\left(r - \rho + \frac{(1+\theta)}{2}\frac{\lambda^2}{\theta^i}\right)}{\theta}dt, \tag{12}$$

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which corresponds to equation (9). Excluding risky asset investments, i.e. $\lambda = 0$, yields equation (5), in which $\phi = \theta$, such that the intertemporal elasticity of substitution of the individual is given by $\frac{1}{\theta}$.

Consumption smoothing

In this section we derive in which way a shock in wealth (dW_t) at time t affects the consumption level (dC_s) at any time $s \geq t$. From equation (4) it follows that:

$$\frac{d\log C_t}{d\log W_t} = 1 \tag{13}$$

since $g(D-t)$ is a deterministic function of time. From equation (11) it follows that consumption growth is independent of wealth such that:

$$\frac{d\log C_s}{d\log C_t} = 1 \tag{14}$$

Combining those two results we obtain equation (10) :

$$\frac{d\log C_s}{d\log W_t} = \frac{d\log C_s}{d\log C_t}\frac{d\log C_t}{d\log W_t} = 1 \tag{15}$$

Technical derivation of expressions in chapter 3

Optimal solution in absence of risk sharing

In absence of risk sharing (i.e. $\tilde{\tau}_s = 0$ for all s) the utility of generation t is given by (substitute (16) in (15) to eliminate C_t)

$$\{U_t\}_{\forall s:\tilde{\tau}_s=0} = E_{t=0}\left[\frac{1}{1-\theta}\left((1+r)Y_t + f_t Y_t \tilde{x}_t\right)\right] \tag{16}$$

The Arrow-Pratt approximation (see e.g. Gollier C. 2001. The Economics of Risk and Time. MIT Press.)

$$\begin{aligned} & E_{t=0}\left[\frac{1}{1-\theta}\left((1+r)Y_t + f_t\tilde{x}_t Y_t\right)^{1-\theta}\right] \\ \approx\ & \mathbf{E}_{t=0}\left[\frac{1}{1-\theta}\left((1+r)Y_t + f_t Y_t\bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t}f_t^2 Y_t^2\sigma^2\right)^{1-\theta}\right] \end{aligned} \tag{17}$$

is applied to obtain (substitute (17) into (16)):

$$\{CEC_t\}_{\forall s:\tilde{\tau}_s=0} \approx (1+r)Y_t + f_t Y_t \bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t} f_t^2 Y_t^2 \sigma^2. \tag{18}$$

in which CEC_t denotes the certainty equivalent consumption level of generation t and is defined as the non-stochastic consumption level that yields U_t:

$$\mathbf{E}_{t=0}\left[\frac{1}{1-\theta}\left(CEC_t\right)^{1-\theta}\right] \equiv U_t \tag{19}$$

In (18), the term $f_t \bar{\mu} Y_t$ represents the excess return on risky investments, and the term $\frac{1}{2}\frac{\theta}{Y_t} f_t^2 Y_t^2 \sigma^2$ represents the welfare loss associated with the volatility in consumption that is due to risk taking. The first-order derivative of (18) with respect to f_t solves the optimal risk exposure:

$$\{f_t\}_{\forall s:\tilde{\tau}_s=0} = \frac{\lambda}{\theta\sigma}. \tag{20}$$

Substitution of (20) in (18) to eliminate f_t yields:

$$\{CEQ_t\}_{\forall s:\tilde{\tau}_s=0} = (1+r)Y_t + \frac{1}{2}\frac{\lambda^2}{\theta} Y_t. \tag{21}$$

in which the second term $\frac{1}{2}\frac{\lambda^2}{\theta} Y_t$ represents the welfare gain associated with the exposure to period-t risk.

Optimal risk sharing solution

In the presence of risk sharing transfers, utility U_t is given by (substitute (16) in (15) to eliminate C_t)

$$U_t = \begin{cases} \mathbf{E}_{t=0}\left[\frac{1}{1-\theta}\left((1+r)Y_t + f_t Y_t \tilde{x}_t - \tilde{\tau}_t\right)^{1-\theta}\right] & \text{for } t=1 \\ \mathbf{E}_{t=0}\left[\frac{1}{1-\theta}\left((1+r)Y_t + f_t Y_t \tilde{x}_t - \tilde{\tau}_t + (1+r)\tilde{\tau}_{t-1}\right)^{1-\theta}\right] & \text{for } t>1 \end{cases}. \tag{22}$$

with $\tilde{\tau}_t$ specified in (17). Applying an Arrow-Pratt approximation (along the lines of (17)) yields the expression for the certainty equivalent consumption CEC_t (defined in (19)):

$$CEC_t = \begin{cases} (1+r)Y_t + (1-\hat{\beta}_t) f_t Y_t \bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t}(1-\hat{\beta}_t)^2 f_t^2 Y_t^2 \sigma^2 - \alpha_t & \text{for } t=1 \\ (1+r)Y_t + (1-\hat{\beta}_t) f_t Y_t \bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t}(1-\hat{\beta}_t)^2 f_t^2 Y_t^2 \sigma^2 - \alpha_t & \\ \quad +(1+r)\left\{\hat{\beta}_{t-1} f_{t-1} Y_{t-1} \bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t}\hat{\beta}_{t-1}^2 f_{t-1}^2 Y_{t-1}^2 \sigma^2 + \alpha_{t-1}\right\} & \text{for } t>1 \end{cases}. \tag{23}$$

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in which $\hat{\beta}_t \equiv \frac{\beta_t}{f_t Y_t}$ represents the fraction of period-t risk that is transferred to generation $t+1$. The remaining fraction $1-\hat{\beta}_t$ is born by generation t.

The optimal risk sharing solution solves from the maximization of the discounted sum of certainty equivalent consumption:

$$\max_{f_t,\alpha_t,\hat{\beta}_t} \sum_{i=1}^{\infty} \frac{CEC_i}{(1+r)^{i-1}}. \tag{24}$$

It follows immediately that the optimal risk sharing policy is unaffected by deterministic transfers α_t (substitution of (23) into (24) to eliminate CEC_t yields an expression that does not depend on α_t).

The optimal allocation of risk across generations solves from substitution of (23) into (24) and taking the first-order derivative with respect to $\hat{\beta}_t$:

$$\hat{\beta}_t = \frac{\eta}{1+\eta} \tag{25}$$

where $\eta \equiv \frac{1}{Y_t}\frac{Y_{t+1}}{1+r}$. Equation (25) states that period-t risk is allocated across generations t and $t+1$ proportional to relative wealth. This leads to consumption smoothing equation (18) in the text (substitute (17) and (4) into (16) to eliminate τ_t and β_t and take the partial derivative with respect to $\tilde{x}_t$).

The optimal risk exposure solves from from substitution of (23) and (25) into (24) and taking the first-order derivative with respect to f_t:

$$f_t = \frac{\lambda}{\theta\sigma}(1+\eta) \tag{26}$$

Equation (19) then follows from subtracting (20) from (26).

Substitution of (25) and (26) into (23) yields:

$$CEC_t = \begin{cases} (1+r)Y_t + \frac{1}{2}\frac{\lambda^2}{\theta}Y_t - \alpha_t & \text{for } t=1 \\ (1+r)Y_t + \frac{1}{2}\frac{\lambda^2}{\theta}Y_t - \alpha_t + \frac{1}{2}\frac{\lambda^2}{\theta}Y_t + (1+r)\alpha_{t-1} & \text{for } t>1 \end{cases}. \tag{27}$$

in which the second term $\frac{1}{2}\frac{\lambda^2}{\theta}Y_t$ in (27) represents the welfare gain associated with the exposure to period-t risk that also appears in (21), and in which the fourth term $\frac{1}{2}\frac{\lambda^2}{\theta}Y_t$ in (27) represents the welfare gain associated with the ability of generation t to share in the risk from period $t-1$ (cf (20)).

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Utility-based fairness criterion

All generations experience the same relative gain from risk sharing in terms of certainty equivalent consumption if:

$$\alpha_t = -\eta \frac{1}{2} \frac{\lambda^2}{\theta} Y_t \tag{28}$$

for all $t \geq 1$, in which case the gain from risk sharing equals (substitute (28) into (27) to eliminate α_t and subsequently subtract (21))

$$CEC_t - \{CEQ_t\}_{\forall s:\tilde{\tau}_s=0} = \eta \frac{1}{2} \frac{\lambda^2}{\theta} Y_t \tag{29}$$

for all $t \geq 1$.

Market-based fairness criterion

In our benchmark model, the ex-ante market value of the intergenerational transfer $\tilde{\tau}_t$ $(t \geq 1)$ is equal to its deterministic component α_t because the loading β_t on the excess return on risky assets has no market value (substitute (21) into (22)):

$$GA_t = \begin{cases} -\alpha_t/(1+r) & \text{for } t = 1 \\ -\alpha_t/(1+r) + \alpha_{t-1} & \text{for } t > 1 \end{cases}. \tag{30}$$

which leads to the zero-sum property in (23). Redistribution in terms of ex-ante market value is ruled out by setting

$$\alpha_t = 0 \tag{31}$$

for all t, in which case $GA_t = 0$ for all t (substitute (31) in (30)). The welfare gain from risk sharing under this criterion is given by (substitute (31) in (27) to eliminate α_t and substract (21))

$$CEC_t - \{CEQ_t\}_{\forall s:\tilde{\tau}_s=0} = \begin{cases} 0 & \text{for } t = 1 \\ \frac{1}{2} \frac{\lambda^2}{\theta} Y_t & \text{for } t > 1 \end{cases}. \tag{32}$$

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Elastic labor supply

Expressions in section 3.4.2 are derived on the basis of a setting in the labor supply of generation $t+1$ (for all $t \geq 1$) is elastic with respect to the size of the transfer $\tilde{\tau}_t$ that is levied upon this generation by the previous generation t. The benchmark model is extended as follows. The labor earnings of generation t are redefined as the product of the wage rate w_t and the labor supply level h_t of generation t, i.e. $Y_t \equiv w_t h_t$. The preferences of generation t, previously given by (15), are now specified over consumption C_t and labor h_t:

$$U_t = \mathbf{E}\left[\frac{1}{1-\theta}\left(C_t - \frac{\epsilon}{\epsilon+1}\left(h_t\right)^{\frac{\epsilon+1}{\epsilon}} + \frac{\epsilon}{\epsilon+1}\left(h_t^*\right)^{\frac{\epsilon+1}{\epsilon}}\right)^{1-\theta}\right], \tag{33}$$

in which ϵ represents the elasticity of labor supply with respect to the marginal wage rate. Accordingly, a drop in the wage rate by one percent results in a decline in the labor supply level of ϵ percent. Originating from Greenwood (1988), the specification in (33) features no income effects in labor supply. If labor supply is undistorted, labor supply choices are given by:

$$h_t^* \equiv w_t^{\epsilon}. \tag{34}$$

and (33) reduces into (15).

Assume that risk-sharing transfer $\tilde{\tau}_{t-1}$ are levied upon the labor earnings of generation t in the form of *proportional* taxes and subsidies.[22] The marginal tax/subsidy on labor earnings then equals the average tax/subsidy and is given by $\frac{\tilde{\tau}_{t-1}}{w_t h_t}$. As a result, the effective marginal wage rate against which labor is supplied by generation t equals $w_t(1+\frac{\tilde{\tau}_{t-1}}{w_t h_t})$ and it follows from (34) that the labor-supply choice h_t of generation t is given by:

$$h_t = \left(w_t\left(1+\frac{\tilde{\tau}_{t-1}}{w_t h_t}\right)\right)^{\epsilon} = h_t^*\left(1+\frac{\tilde{\tau}_{t-1}}{w_t h_t}\right)^{\epsilon} \approx h_t^*\left(1+\frac{\tilde{\tau}_{t-1}}{w_t h_t^*}\right)^{\epsilon}, \tag{35}$$

where the approximation is applied because the labor supply choice h_t appears on both sides of the first equality of (35) and does not adopt an explicit solution.

[22] Proportional taxes and subsidies are, by approximation, optimal if preferences over consumption obey constant relative risk aversion. Under constant relative risk aversion, it is optimal for agents to share proportionally equally in a shocks. This result holds exactly in absence of distortions, and holds by approximation in the model with elastic labor supply if distortions are not too large.

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Under elastic labor supply, the deterministic component α_t matters for the social surplus from risk sharing because it distorts labor supply. We therefore derive expressions for the welfare gain from risk sharing under the condition $\alpha_t = 0$ as in (31). Substitution of (35) into (33) and applying an Arrow-Pratt approximation along the lines of (18) and (23), the expression for certainty equivalent utility now becomes:

$$CEC_t = \begin{cases} (1+r)Y_t^* + (1-\hat{\beta}_t)f_t Y_t^* \bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t^*}(1-\hat{\beta}_t)^2 f_t^2 (Y_t^*)^2\sigma^2 - \alpha_t & \text{for } t=1 \\ (1+r)Y_t^* + (1-\hat{\beta}_t)f_t Y_t^* \bar{\mu} - \frac{1}{2}\frac{\theta}{Y_t^*}(1-\hat{\beta}_t)^2 f_t^2 (Y_t^*)^2\sigma^2 - \alpha_t & \\ \quad +(1+r)\left\{\hat{\beta}_{t-1} f_{t-1} Y_{t-1}^* \bar{\mu} - \frac{1}{2}\frac{\theta+\epsilon}{\theta}\frac{\theta}{Y_t^*}\hat{\beta}_{t-1}^2 f_{t-1}^2 (Y_{t-1}^*)^2\sigma^2 + \alpha_{t-1}\right\} & \text{for } t>1 \end{cases}. \tag{36}$$

in which $Y_t^* \equiv w_t h_t^*$. The term $\frac{1}{2}\frac{\theta+\epsilon}{Y_t^*}\hat{\beta}_{t-1}^2 f_{t-1}^2 (Y_{t-1}^*)^2\sigma^2$ now includes not only the welfare loss from volatility in consumption associated with the risk transfer (as measured by θ) but now also includes the welfare loss from from labor-supply distortions associated with the risk transfer (as measured by ϵ). In the special case where labor supply is inelastic, i.e. $\epsilon = 0$, certainty equivalent consumption in (36) reduces into (23).

Substituting (36) into (24) and taking the first-order conditions with respect to $\hat{\beta}_t$ and f_t solves the optimal risk sharing solution under elastic labor supply (cf (25) and (26)):

$$\hat{\beta}_t = \frac{\eta}{1+\eta}\frac{\theta}{\theta+\epsilon}. \tag{37}$$

and

$$f_t = \frac{\lambda}{\theta\sigma}\left(1+\eta\frac{\theta}{\theta+\epsilon}\right), \tag{38}$$

where now $\eta \equiv \frac{1}{Y_t^*}\frac{Y_{t+1}^*}{1+r}$. (37) leads to (18) in the same way as (37) leads to (18).

Substitution of (37) and (38) into (36) yields: (cf (27))

$$CEC_t = \begin{cases} (1+r)Y_t + \frac{1}{2}\frac{\lambda^2}{\theta}Y_t - \alpha_t & \text{for } t=1 \\ (1+r)Y_t + \frac{1}{2}\frac{\lambda^2}{\theta}Y_t - \alpha_t + \frac{1}{2}\frac{\lambda^2}{\theta}\frac{\theta}{\theta+\epsilon}Y_t + (1+r)\alpha_{t-1} & \text{for } t>1 \end{cases}. \tag{39}$$

in which the term $\frac{1}{2}\frac{\lambda^2}{\theta}\frac{\theta}{\theta+\epsilon}Y_t$ represents the welfare gain associated with the ability of generation t to share in the risk from period $t-1$ (cf (25)).